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CREATE TABLE Departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(100) UNIQUE NOT NULL,
    dept_location VARCHAR(100)
);

CREATE TABLE Employees (
    emp_id INT PRIMARY KEY,
    emp_firstname VARCHAR(50) NOT NULL,
    emp_lastname VARCHAR(50) NOT NULL,
    emp_email VARCHAR(100) UNIQUE,
    phn_no CHAR(10),
    salary DECIMAL(10, 2) CHECK (Salary > 0),
    hiring_date DATE,
    gender CHAR(1) CHECK (Gender IN ('M', 'F')),
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)
);

DESC Departments;
DESC Employees;

INSERT INTO Departments VALUES
(1, 'CSE', 'Building V'),
(2, 'CSE-AI/ML', 'Building VI'),
(3, 'CSE-DS', 'Building VII');

INSERT INTO Employees VALUES
(101, 'Ram', 'Das', 'ram@gmail.com', 9876543210, 50000.25, CURDATE(), 'M', 1),
(102, 'Sham', 'Roy', 'sham@gmail.com', 9876883210, 60000.25, CURDATE(), 'M', 2),
(103, 'Kalyan', 'Biswas', 'kalyan@gmail.com', 9899543210, 86000.25, CURDATE(), 'M', 3),
(104, 'Sayani', 'Paul', 'sayani@gmail.com', 9876544210, 38000.25, CURDATE(), 'F', 1),
(105, 'Prem', 'Shau', 'prem@gmail.com', 9866643210, 25000.25, CURDATE(), 'M', 3);

-- 1. Display all records from the Employees table.
SELECT * FROM Employees;

-- 2. Display only the FirstName and Salary of all employees.
SELECT emp_firstname, salary FROM Employees;

-- 3. Display the details of employees whose salary is greater than 40,000.
SELECT * FROM Employees WHERE salary > 40000;

-- 4. Update the salary of the employee with EmployeeID = 101 to 60,000.
UPDATE Employees SET salary = 60000 WHERE emp_id = 101;
SELECT * FROM Employees;

-- 5. Update the salary to 65,000 and phone number to 9999999999 for the employee whose email is amit.verma@example.com.
UPDATE Employees SET emp_email = 'amit.verma@example.com' WHERE emp_id = 101;
UPDATE Employees SET salary = 65000, phn_no = 9988776655 WHERE emp_email = 'amit.verma@example.com';
SELECT * FROM Employees;

-- 6. Delete the employee whose EmployeeID = 101.
DELETE FROM Employees WHERE emp_id = 101;
SELECT * FROM Employees;

-- 7. Delete all employees belonging to Department ID = 1.
DELETE FROM Employees WHERE dept_id = 1;
SELECT * FROM Employees;

-- 8. Remove all records from the Employees table using the TRUNCATE statement.
TRUNCATE TABLE Employees;
DESC Employees;

-- 9. Drop the Employees table.
DROP TABLE Employees;

-- 10. Drop the CompanyDB database. (skipping)

-- 11. Add a new column named ManagerName (VARCHAR(100)) to the Departments table.
ALTER TABLE Departments ADD manager_name VARCHAR(100);
DESC Departments;

-- 12. Modify the Salary column in the Employees table to DECIMAL(12,2).
ALTER TABLE Employees MODIFY salary DECIMAL(12,2);
DESC Employees;

-- 13. Rename the ManagerName column to HeadOfDepartment.
ALTER TABLE Departments RENAME COLUMN manager_name To head_of_department;
DESC Departments;

-- 14. Drop the HeadOfDepartment column from the Departments table.
ALTER TABLE Departments DROP head_of_department;
DESC Departments;

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#	Time	Action	Message
17	13:47:00	SELECT * FROM Employees LIMIT 0, 1000	5 row(s) returned
18	13:47:00	DELETE FROM Employees WHERE emp_id=101	1 row(s) affected
19	13:47:00	SELECT * FROM Employees LIMIT 0, 1000	4 row(s) returned
20	13:47:00	DELETE FROM Employees WHERE dept_id=1	1 row(s) affected
21	13:47:00	SELECT * FROM Employees LIMIT 0, 1000	3 row(s) returned
22	13:47:00	ALTER TABLE Departments ADD manager_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
23	13:47:00	DESC Departments	4 row(s) returned
24	13:47:00	ALTER TABLE Employees MODIFY salary DECIMAL(12,2)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
25	13:47:00	DESC Employees	9 row(s) returned
26	13:47:00	ALTER TABLE Departments RENAME COLUMN manager_name To head_of_department	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
27	13:47:00	DESC Departments	4 row(s) returned
28	13:47:00	ALTER TABLE Departments DROP head_of_department	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
29	13:47:00	DESC Departments	3 row(s) returned
30	13:47:00	TRUNCATE TABLE Employees	0 row(s) affected
31	13:47:00	DESC Employees	9 row(s) returned
32	13:47:00	DROP TABLE Employees	0 row(s) affected

Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	
emp_firstname	varchar(50)	NO		NULL	
emp_lastname	varchar(50)	NO		NULL	
emp_email	varchar(100)	YES	UNI	NULL	
phn_no	char(10)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
hiring_date	date	YES		NULL	
gender	char(1)	YES		NULL	
dept_id	int	YES	MUL	NULL	

department schema × employee schema all records in employee

emp_id	emp_firstname	emp_lastname	emp_email	phn_no	salary	hiring_date	gender	dept_id
101	Ram	Das	ram@gmail.com	9876543210	50000.25	2026-08-20	M	1
102	Sham	Roy	sham@gmail.com	9876883210	60000.25	2026-08-20	M	2
103	Kalyan	Biswas	kalyan@gmail.com	9899543210	86000.25	2026-08-20	M	3
104	Sayani	Paul	sayani@gmail.com	9876544210	38000.25	2026-08-20	F	1
105	Prem	Shau	prem@gmail.com	9866643210	25000.25	2026-08-20	M	3
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

department schema employee schema all records in employee × employee and salary salary more than 40000

emp_id	emp_firstname	emp_lastname	emp_email	phn_no	salary	hiring_date	gender	dept_id
101	Ram	Das	ram@gmail.com	9876543210	50000.25	2026-08-20	M	1
102	Sham	Roy	sham@gmail.com	9876883210	60000.25	2026-08-20	M	2
103	Kalyan	Biswas	kalyan@gmail.com	9899543210	86000.25	2026-08-20	M	3
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

salary more than 40000 × updated salary update email amd phone number delete employee 101 deleting dept wit

emp_id	emp_firstname	emp_lastname	emp_email	phn_no	salary	hiring_date	gender	dept_id
101	Ram	Das	ram@gmail.com	9876543210	60000.00	2026-08-20	M	1
102	Sham	Roy	sham@gmail.com	9876883210	60000.25	2026-08-20	M	2
103	Kalyan	Biswas	kalyan@gmail.com	9899543210	86000.25	2026-08-20	M	3
104	Sayani	Paul	sayani@gmail.com	9876544210	38000.25	2026-08-20	F	1
105	Prem	Shau	prem@gmail.com	9866643210	25000.25	2026-08-20	M	3
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

updated salary × update email amd phone number delete employee 101 deleting dept with id adding a new column

Output

Result Grid									
	emp_id	emp_firstname	emp_lastname	emp_email	phn_no	salary	hiring_date	gender	dept_id
▶	101	Ram	Das	amit.verma@example.com	9988776655	65000.00	2026-08-20	M	1
	102	Sham	Roy	sham@gmail.com	9876883210	60000.25	2026-08-20	M	2
	103	Kalyan	Biswas	kalyan@gmail.com	9899543210	86000.25	2026-08-20	M	3
	104	Sayani	Paul	sayani@gmail.com	9876544210	38000.25	2026-08-20	F	1
	105	Prem	Shau	prem@gmail.com	9866643210	25000.25	2026-08-20	M	3
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

update email amd phone number x delete employee 101 deleting dept with id adding a new column modifying salary re

Output

Result Grid									
	emp_id	emp_firstname	emp_lastname	emp_email	phn_no	salary	hiring_date	gender	dept_id
▶	102	Sham	Roy	sham@gmail.com	9876883210	60000.25	2026-08-20	M	2
	103	Kalyan	Biswas	kalyan@gmail.com	9899543210	86000.25	2026-08-20	M	3
	104	Sayani	Paul	sayani@gmail.com	9876544210	38000.25	2026-08-20	F	1
	105	Prem	Shau	prem@gmail.com	9866643210	25000.25	2026-08-20	M	3
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

delete employee 101 x deleting dept with id adding a new column modifying salary renaming table dropping col

Result Grid									
	emp_id	emp_firstname	emp_lastname	emp_email	phn_no	salary	hiring_date	gender	dept_id
▶	102	Sham	Roy	sham@gmail.com	9876883210	60000.25	2026-08-20	M	2
	103	Kalyan	Biswas	kalyan@gmail.com	9899543210	86000.25	2026-08-20	M	3
	105	Prem	Shau	prem@gmail.com	9866643210	25000.25	2026-08-20	M	3
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

deleting dept with id x adding a new column modifying salary renaming table dropping column truncating table

Result Grid									
	Field	Type	Null	Key	Default	Extra			
▶	dept_id	int	NO	PRI	NULL				
	dept_name	varchar(100)	NO	UNI	NULL				
	dept_location	varchar(100)	YES		NULL				
	manager_name	varchar(100)	YES		NULL				

Result Grid									
	Field	Type	Null	Key	Default	Extra			
▶	emp_id	int	NO	PRI	NULL				
	emp_firstname	varchar(50)	NO		NULL				
	emp_lastname	varchar(50)	NO		NULL				
	emp_email	varchar(100)	YES	UNI	NULL				
	phn_no	char(10)	YES		NULL				
	salary	decimal(12,2)	YES		NULL				
	hiring_date	date	YES		NULL				
	gender	char(1)	YES		NULL				
	dept_id	int	YES	MUL	NULL				

adding a new column x modifying salary renaming table dropping column

modifying salary x renaming table dropping column truncating table

Output

Result Grid									
	Field	Type	Null	Key	Default	Extra			
▶	dept_id	int	NO	PRI	NULL				
	dept_name	varchar(100)	NO	UNI	NULL				
	dept_location	varchar(100)	YES		NULL				
	head_of_department	varchar(100)	YES		NULL				

renaming table x dropping column truncating table

Output

Action Output

Result Grid									
	Field	Type	Null	Key	Default	Extra			
▶	dept_id	int	NO	PRI	NULL				
	dept_name	varchar(100)	NO	UNI	NULL				
	dept_location	varchar(100)	YES		NULL				

dropping column x truncating table

Result Grid									
	Field	Type	Null	Key	Default	Extra			
	emp_id	int	NO	PRI	NULL				
	emp_firstname	varchar(50)	NO		NULL				
	emp_lastname	varchar(50)	NO		NULL				
	emp_email	varchar(100)	YES	UNI	NULL				
	phn_no	char(10)	YES		NULL				
▶	salary	decimal(12,2)	YES		NULL				
	hiring_date	date	YES		NULL				
	gender	char(1)	YES		NULL				
	dept_id	int	YES	MUL	NULL				

truncating table x